Domain and Range

Domain from equations

1 State the domain of each function:

a $f(x) = \frac{1}{x-4}$

b $g(x) = \sqrt{2x-6}$

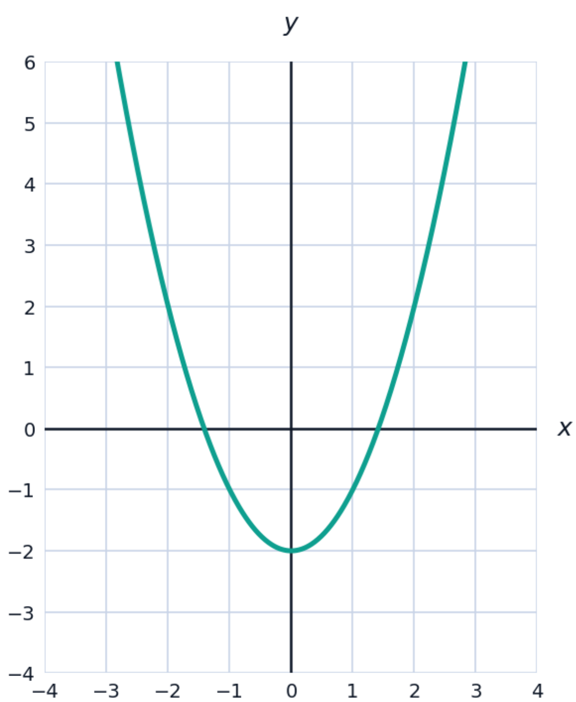
c $h(x) = \frac{x+1}{x^2-9}$

d $k(x) = \sqrt[3]{x}$

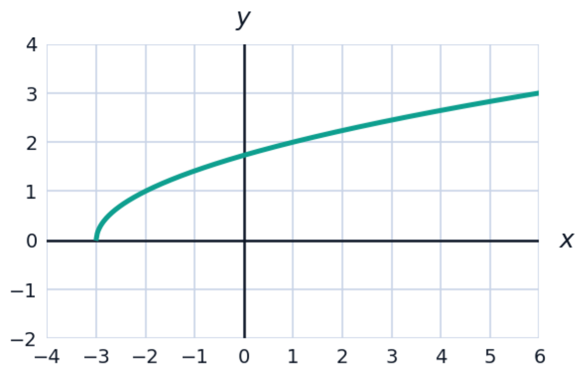
2 Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.

Domain and range from graphs

3 The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.



4 The graph of $y = \sqrt{x+3}$ is shown. State the domain and range of the function.



Finding the range algebraically

5 By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

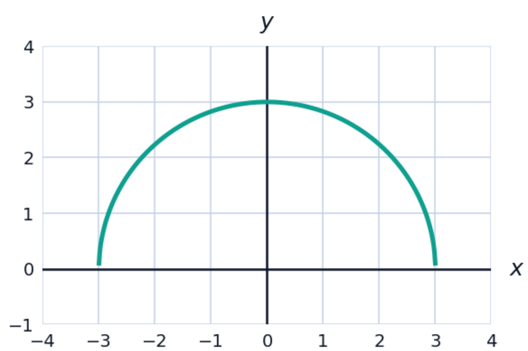
6 Consider $f(x) = -2(x-1)^2 + 5$.

a State the coordinates of the vertex.

b Does the parabola open upward or downward?

c State the range of f.

7 Find the range of $g(x) = \sqrt{9 - x^2}$ (the upper half of a circle).



Domain and range in applied contexts

8 A rectangular garden has perimeter 40 m. If x is the width, the length is $20 - x$, and the area is $A(x) = x(20-x)$.

a State the domain of A in this context (both dimensions must be positive).

b Find $A(6)$ and $A(15)$.

c State the range of A over this domain, to the nearest whole number.

9 An open-top box is formed from a 12 cm square by cutting squares of side x from each corner, giving volume $V(x) = x(12-2x)^2$.

a State the domain of V in this physical context.

b Find $V(1)$ and $V(4)$.

Domain of composite and combined functions

10 Let $f(x) = \sqrt{x-2}$ and $g(x) = \frac{1}{x}$. Find the domain of:

a $(f+g)(x)$

b $(f \circ g)(x)$

11 Find the domain of $f(x) = \frac{1}{\sqrt{x^2-4}}$, writing your answer in interval notation.

Domain from rational functions with factoring

12 Find the domain of $f(x) = \frac{x^2-4}{x^2-x-6}$, factoring the denominator first.

13 Find the domain of $g(x) = \frac{\sqrt{x+4}}{x-1}$, combining a root restriction with a denominator restriction.

Range using transformations

14 The range of $f(x) = x^2$ is $y \geq 0$. State the range of $g(x) = f(x) - 5 = x^2 - 5$ and of $h(x) = -f(x) = -x^2$, using transformation reasoning.

15 The range of $f(x) = \sqrt{x}$ is $y \geq 0$. State the range of $g(x) = 3\sqrt{x} + 2$.

Domain and range of piecewise functions

16 A function is defined by $f(x) = x^2$ for $x < 0$ and $f(x) = 2x + 1$ for $x \geq 0$. State the domain and range.